

BUS63-7

# Entrepreneurial Finance

Startup financial lifecycle, financial modeling, unit economics, cash dynamics, the investor ecosystem, term sheets, cap tables, exit waterfalls, and the founder financial pitch

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The Finance arc applied from the founder's chair: cap tables, term sheets, and the documents that decide founder outcomes.

Albert School — 22 hours

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## 1 Financial lifecycle of a startup

This course applies the Core Finance arc (BUS13-3  $\rightarrow$  BUS43-3) — the three statements, ratios, free cash flow, NPV, WACC, DCF, and multiples — to the venture-backed context, reasoning from the founder’s perspective rather than the analyst’s or CFO’s.

**Definition 1** The **financial lifecycle** of a startup is the sequence of stages — pre-launch, launch, growth, break-even, scale — through which its cash needs, revenue patterns, and financing options evolve. Each stage has a characteristic financial profile: pre-revenue burn financed by equity, accelerating revenue with continued investment, and eventual self-funding at scale.

**Definition 2** A **financial inflection point** is a stage transition at which the company’s financial logic changes — first revenue, first positive contribution margin, cash-flow break-even, and the points at which a new financing round becomes necessary or possible.

### Exercises

1. Describe how cash needs, revenue patterns, and financing options evolve for a startup from pre-launch through break-even to scale.
2. Identify the critical financial inflection points at each stage of the lifecycle.

## 2 Startup financial modeling

**Definition 3** A **bottoms-up driver model** projects revenue from operational drivers (e.g. number of customers, conversion rate, average price, usage) rather than from a top-down market-share assumption. Each revenue line is the product of explicit, defensible driver assumptions.

**Definition 4** A **startup financial model** projects, over an 18-to-24-month horizon, revenues (bottoms-up), operating costs and headcount, and a **monthly cash flow statement**. The model maintains **coherence** across the three views: the P&L, the cash flow statement,

and the cap table must reconcile — the same financing events and the same results appear consistently in each.

**Common pitfall** Monthly granularity is essential for a startup model: a model built on annual figures hides the months in which cash runs out. Revenue and cash are not the same line — a sale recognised in the P&L may be collected in cash later, and the model must track both.

#### Exercises

3. Build a startup financial model from scratch: project revenues with a bottoms-up driver model, forecast operating costs and headcount, and derive a monthly cash flow statement over an 18-to-24-month horizon.
4. Demonstrate P&L / cash flow / cap-table coherence in the model.

## 3 Unit economics

**Definition 5 Unit economics** describe the profitability of a single unit of activity (one customer or one transaction):

- **Contribution margin** — revenue per unit minus variable cost per unit;
- **Gross margin** — revenue minus cost of goods sold, as a share of revenue;
- **CAC** (Customer Acquisition Cost) — total sales-and-marketing spend divided by the number of customers acquired;
- **LTV** (Lifetime Value) — the total contribution a customer generates over the expected lifetime of the relationship;
- **Payback period** — the time required for a customer's cumulative contribution to recover the CAC.

**Definition 6 Viability thresholds** are the levels these metrics must reach for the business model to be sustainable — notably an LTV/CAC ratio comfortably above 1 and a CAC payback period short relative to customer lifetime. The thresholds depend on the business model.

#### Exercises

5. Calculate contribution margin, CAC, LTV, payback period, and gross margin for a business.
6. Explain what each metric reveals about the sustainability of the business model, and state the viability thresholds that apply.

## 4 Cash dynamics: burn, runway, and break-even

**Definition 7 Burn rate** is the rate at which a startup consumes cash.

- **Gross burn** is total monthly cash outflow (operating expenses paid).

- **Net burn** is monthly cash outflow minus cash inflow (net cash consumed):

$$\text{net burn} = \text{cash out} - \text{cash in}.$$

**Definition 8** **Runway** is the number of months the company can operate before exhausting its cash, at the current net burn:

$$\text{runway} = \frac{\text{cash on hand}}{\text{net monthly burn}}.$$

**Definition 9** The **break-even point** is the level of activity at which total revenue equals total cost, so profit is zero. From a cost structure split into fixed costs  $F$  and a per-unit contribution margin  $m$  (price minus variable cost per unit), the break-even volume is

$$Q^* = \frac{F}{m}.$$

Pricing, volume, and variable-cost assumptions each shift  $Q^*$ .

**Common pitfall** The levers that extend runway — slowing hiring, cutting spend, raising prices — can also slow growth. The founder's task is to extend survival **without killing growth**: identify the cuts that buy the most months per unit of growth sacrificed.

### Exercises

7. Calculate gross burn, net burn, and runway for a startup, and explain the relationship between hiring pace, revenue growth, and cash exhaustion.
8. Identify the levers a founder can pull to extend runway without killing growth.
9. Calculate the break-even point from a cost-structure model and describe how pricing, volume, and variable-cost assumptions shift it.

## 5 The investor ecosystem

**Definition 10** The **early-stage funding landscape** comprises:

- **Bootstrapping** — financing growth from revenue and founders' own funds, with no external equity;
- **Business angels** — individual or institutional private investors writing early cheques;
- **Accelerators** — programmes providing small capital, mentoring, and network in exchange for equity;
- **Seed funds** — institutional funds specialised in the first priced rounds.

**Definition 11** **Venture capital fund economics**: a VC fund is raised from **Limited Partners** (LPs, the capital providers) and managed by **General Partners** (GPs). GPs earn a **management fee** and **carried interest** — a share (typically around 20%) of the fund's profits above the returned capital. A fund's **vintage** is the year it begins investing;

returns are measured by vintage cohort. VCs specialise by stage — **early-stage**, **growth**, and **corporate VC** — each with a distinct return profile.

**Definition 12 Private equity**, by contrast with VC, deploys larger cheques into more mature companies: **buyout** (acquiring control, often with leverage) and **growth equity** (minority stakes in established, growing companies).

### Exercises

10. Describe the startup fundraising landscape (bootstrapping, angels, accelerators, seed funds) and venture capital fund economics (LP/GP, carried interest, vintages, stage specialisation).

11. Select investors for a venture based on criteria beyond the cheque.

## 6 Term sheets clause by clause

**Definition 13** Three instruments fund early rounds:

- a **SAFE** (Simple Agreement for Future Equity) — a right to equity in a future priced round, with no debt features;
- a **convertible note** — debt that converts to equity at a later round, carrying interest and a maturity date;
- a **priced equity round** — an immediate sale of shares at an agreed per-share price.

SAFEs and notes typically carry a **valuation cap** and/or **discount** that set the conversion price, with differing dilution and investor-rights implications.

**Definition 14** A **liquidation preference** entitles preferred investors to be paid before common holders at exit. It is:

- **non-participating** — the investor takes the greater of its preference or its as-converted share, but not both;
- **participating** — the investor takes its preference **and** then also shares in the remainder pro rata (“double dip”).

The preference **multiple** ( $1\times$ ,  $2\times$ , ...) scales the amount returned first.

**Definition 15** **Anti-dilution** protection adjusts preferred investors’ conversion price in a **down round**:

- **broad-based weighted average** — adjusts partially, accounting for the size of the new issuance relative to the existing share base;
- **full ratchet** — repriced to the new, lower price regardless of issuance size, the more founder-punitive form.

**Definition 16** **Governance and exit clauses** allocate control and shape exits:

- **pro-rata rights** — the right to invest in future rounds to maintain ownership percentage;

- **drag-along** — majority can compel minority to join a sale;
- **tag-along (co-sale)** — minority can join a sale on the same terms;
- **right of first refusal (ROFR)** — the company or existing investors may match a third-party offer for shares before they are sold.

**Common pitfall** The headline valuation is not the deal. A high pre-money paired with a participating 2× liquidation preference and full-ratchet anti-dilution can leave the founder worse off at exit than a lower valuation with clean terms. Each clause must be quantified for its impact on the founder's outcome, not read in isolation.

### Exercises

12. Read a term sheet clause by clause and quantify the impact of liquidation preference (participating vs non-participating), anti-dilution (broad-based weighted average vs full ratchet), pro-rata, and governance/exit clauses on the founder's outcome.
13. Distinguish SAFE, convertible notes, and priced equity rounds and explain their differing dilution and investor-rights implications.

## 7 Cap-table construction and round mechanics

**Definition 17** A **capitalisation table** (cap table) records who owns what: the number of shares and the ownership percentage of each holder — **founders**, **BSPCEs** (employee stock options / warrants), and successive **investors**.

**Definition 18** **Pre-money vs post-money:**

- **pre-money valuation** is the agreed value of the company before the new investment;
- **post-money valuation** = pre-money + amount invested.

The investor's ownership is

$$\text{investor \%} = \frac{\text{amount invested}}{\text{post-money valuation}},$$

and existing holders are **diluted** in proportion.

**Definition 19** **Cap-table evolution** across **seed**, **Series A**, and **Series B** compounds dilution: each round issues new shares, reducing every prior holder's percentage. The **cumulative dilution** on founders, employees, and earlier investors is the product of the retained fractions across all rounds.

**Definition 20** At each stage investors assess **traction metrics** appropriate to that stage (e.g. early product signals at seed, revenue growth and retention at Series A, efficient scaling at Series B). The metrics that justify the next round rise with each stage.

### Exercises

14. Describe the Series A/B fundraising process and the traction metrics investors use at each stage.
15. Model cap-table evolution (pre-money vs post-money mechanics) and the cumulative dilution impact of successive rounds on founders, employees, and earlier investors.

## 8 Common founder financial mistakes

**Common pitfall** The financial mistakes founders most commonly make in their first two years:

- **over-hiring ahead of revenue** — scaling headcount before the revenue to support it, shortening runway;
- **mispricing** — setting prices that fail to cover unit costs or to reflect value;
- **ignoring unit economics** — growing a model that loses money on every unit;
- **conflating revenue growth with cash health** — treating top-line growth as proof of survival while cash is being exhausted.

Each is avoided by the discipline taught in this course: monthly cash modelling, unit-economics checks before scaling, and pricing tied to contribution margin.

### Exercises

16. Identify the common financial mistakes founders make in their first two years and explain how each can be avoided.

## 9 Alternative financing

**Definition 21** Alternative (non-equity) financing instruments:

- **venture debt** — a loan to a venture-backed company, repaid with interest (often alongside warrants), that extends runway without immediate dilution;
- **revenue-based financing** — capital repaid as a fixed percentage of future revenue until a capped multiple is reached.

Each carries an **effective cost** — the all-in cost of the capital including interest, fees, and any warrants — that can be compared with the dilution cost of equity.

**Definition 22** Appropriateness depends on the **revenue profile** and **risk tolerance**: debt and revenue-based financing suit companies with predictable revenue able to service repayment; equity suits pre-revenue or highly uncertain stages where fixed repayment would threaten survival.

### Exercises

17. Compare venture debt and revenue-based financing to equity: calculate the effective cost of each.
18. Explain when each is appropriate given a company's revenue profile and risk tolerance.

## 10 The exit waterfall

**Definition 23** An **exit waterfall** allocates the proceeds of an exit across the cap table in order of priority: liquidation preferences are paid first (in their seniority order), then the remainder is distributed to common holders — with **participating** preferred also sharing in that remainder. The payout to each holder is computed by running the proceeds down this priority stack.

**Definition 24** The waterfall is evaluated at multiple **exit valuations** — typically  $1\times$ ,  $3\times$ , and  $10\times$  the entry valuation — because the distribution of value between investors and founders/employees changes with the exit size. At low exits liquidation preferences dominate and founders may receive little; at high exits preferences become a small fraction and value flows to common holders.

**Common pitfall** Cap-table structure shapes **incentive alignment** at exit. Heavy liquidation preferences can leave founders and employees with nothing in a modest exit, removing their incentive to accept it — so the structure agreed at financing determines whose interests align at sale.

### Exercises

19. Model a startup exit across a cap-table waterfall at  $1\times$ ,  $3\times$ , and  $10\times$  entry valuations: calculate founder, employee, and investor payouts under different liquidation-preference and participation scenarios.
20. Explain how cap-table structure shapes incentive alignment at exit.

## 11 Founder-side due diligence and the financial pitch

**Definition 25** **Founder-side due diligence** is the founder's own diagnosis of the company's financial statements (real or simulated) — cash position, burn, runway, and risks — in anticipation of investor scrutiny.

**Definition 26** The **financial pitch** defends the model before an investor audience. **Three slides** differentiate a serious pitch:

- **unit economics** — that each unit is or becomes profitable;
- **runway** — how long the cash lasts and what the raise buys;
- **use of funds** — where the money goes and what milestone it reaches.

The discipline is **defending assumptions** under due-diligence questioning in real time.

### Exercises

21. Analyse the financial statements of a real or simulated early-stage company to diagnose its cash position and risks.



**22.** Pitch the resulting financial model to an investor audience, defending unit-economics assumptions and responding to due-diligence questions in real time.

This course completes the Entrepreneurship-track financial toolkit, pivoting the Core Finance arc's NPV, WACC, DCF, and multiples onto the documents — cap tables, term sheets, and VC memos — that decide founder outcomes at the negotiating table.